# Unified Cognition from a Single Optimization Principle: Active Inference in MeTTa with Emergent Reasoning, Planning, and Self-Knowledge

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March 2026

**Abstract**

We present a cognitive architecture targeting artificial general intelligence (AGI)—in the sense of Goertzel and Pennachin [2007]: domain-general, transfer-capable, and representation-learning—in which perception, action selection, causal reasoning, temporal planning, self-knowledge, and conversation all derive from a single optimization principle: minimize Expected Free Energy (EFE). The architecture is implemented in approximately 12,000 lines of MeTTa, a symbolic meta-language for knowledge representation and reasoning, and validated at three levels: executable Python specifications encoding identical logic; a pure Python MeTTa evaluator that loads and executes the canonical MeTTa source through generic pattern matching; and a stateless LLM orchestrator that runs the full perception–cognition–verbalization pipeline end to end with a local language model. Three forms of inference—induction, deduction, and abduction—operate as hypothesis generators under a uniform metabolic selection mechanism: hypotheses that predict correctly earn energy; those that do not are removed. This yields emergent behaviors including spontaneous investigation of hidden causes (the "Sherlock Holmes effect"), principled retreat under viability threat, and honest uncertainty grounded in actual query failure rather than engineered responses. Temporal planning employs an adaptive beam search inspired by Renormalization Group flow, reducing planning complexity from $O(|A|^d)$ to $O(|A| \times k \times d)$ and achieving an 80,000× reduction in evaluations at 50 actions. A magnitude-banded covariance pruning scheme for the Hebbian suspicion accumulator reduces pairwise scaling from $O(E^2)$ to $O(E \cdot b)$, achieving a 19.4× pair reduction at 1,000 observables with zero significant false negatives. An integrated evaluation on a 14-observable coral reef monitoring scenario demonstrates causal discovery of four hidden chains, metabolic pruning of incorrect models, and—via ablation—that abductive inference is a necessary component for investigation behavior. A boundary condition fix—near-zero initial precision and metabolic capital for abduction—resolves the integration challenge of distinguishing ignorance from anomaly without introducing thresholds, producing 100% Sherlock Holmes effect across all parameter configurations tested.

**Keywords:** Active Inference · Expected Free Energy · Artificial General Intelligence · Cognitive Architecture · MeTTa · Abductive Reasoning · Structure Learning · Metabolic Selection · Magnitude-Banded Pruning

**Code:** https://github.com/Jolnir-Sefir/dagaz
**DOI:** https://doi.org/10.6084/m9.figshare.31742059

## 1 Introduction

Active Inference [Friston et al., 2017, Parr et al., 2022] proposes that adaptive agents can be understood as minimizing a single quantity—the Expected Free Energy (EFE)—which unifies

perception, action, learning, and exploration under one variational principle. This framework has attracted significant theoretical attention, yet most implementations either remain at the level of mathematical formalism, rely on neural network approximations that sacrifice transparency [Millidge, 2020, Catal et al., 2020], or are restricted to narrow domains with hand-crafted generative models.

We present **Project Dagaz**, a cognitive architecture that takes Active Inference seriously as a computational principle. We use *artificial general intelligence* (AGI) in the sense defined by Goertzel and Pennachin [2007]: a system capable of handling cognitive tasks across domains it was not specifically engineered for, transferring learned structure between contexts, and acquiring novel representations from experience. The system is implemented in MeTTa [Potapov, 2021], a symbolic meta-language designed for knowledge representation and reasoning within the OpenCog Hyperon framework [Goertzel et al., 2014], and validated at three levels: executable Python specifications that encode identical dynamics, a pure Python MeTTa evaluator that executes the canonical source, and an end-to-end orchestrator that runs the full cognitive pipeline with a local language model (Section 4). Every inference is traceable, every belief is queryable, and every action selection decision can be decomposed into its contributing terms.

The core claim of this paper is that a single optimization principle—minimize EFE—combined with a uniform metabolic selection mechanism over symbolically generated hypotheses, produces a surprisingly rich set of cognitive behaviors without scripted rules or mode switches. Specifically, we demonstrate:

1. **Unified action selection.** Exploration, exploitation, and self-preservation emerge from the same EFE equation, with no thresholds, urgency levels, or mode flags.

2. **Emergent causal reasoning.** Three forms of inference—induction, deduction, and abduction—are implemented as hypothesis generators feeding a single metabolic selection mechanism. Wrong hypotheses die; correct ones persist.

3. **The Sherlock Holmes effect.** Abductive inference hypothesizes hidden causes, which creates low-precision beliefs, which in turn makes confirming observations the highest information-gain action. The system spontaneously investigates—no module was programmed to produce this behavior.

4. **Efficient temporal planning.** An adaptive beam search inspired by Renormalization Group (RG) flow coarse-grains the planning tree, achieving up to 80,000× reduction in evaluations at 50 actions while preserving correct action selection across all random seeds tested.

5. **Honest self-knowledge.** The system tracks what it can and cannot explain through semantic grounding chains. "I don't know" emerges from actual grounding failure, not from an engineered response.

6. **Scalable causal discovery.** A magnitude-banded covariance pruning scheme reduces the Hebbian suspicion accumulator's pairwise cost from $O(E^2)$ to $O(E \cdot b)$, with the pruning threshold derived from the same metabolic parameters that govern hypothesis selection. At 1,000 observables, this achieves a 19.4× pair reduction with zero significant false negatives—causal discovery is preserved exactly.

**Development Methodology.** This architecture was developed through human-AI collaboration over approximately four months, with the author serving as systems architect and a large language model (Claude, Anthropic) as the primary tool for code generation, mathematical research, and benchmark development. Architectural decisions, design principles, module decomposition, and iterative refinement—including the principled rejection of approaches that

violated design constraints (e.g., replacing epistemic gates with boundary conditions)—were human-directed. Implementation of the MeTTa modules, Python executable specifications, and supporting analyses was AI-assisted. All code and specifications in the repository were reviewed and validated by the author. This methodology is disclosed not as a caveat but as a factual description: the intellectual contribution of this work lies in the architecture—the choice of invariants, the identification of symmetries, and the design constraints that shaped development—rather than in the mechanical production of source code.

The remainder of this paper is organized as follows. Section 2 provides background on Active Inference, Peircean reasoning, and MeTTa. Section 3 describes the architecture. Section 4 presents benchmark results. Section 5 discusses limitations, related work, and implications. Section 6 concludes.

# 2 Background

## 2.1 Active Inference and Expected Free Energy

Active Inference [Friston et al., 2017, Parr et al., 2022] casts perception and action as a single process of free energy minimization. An agent maintains a generative model of its environment and acts to minimize the discrepancy between its predictions and its observations. The Expected Free Energy (EFE) of a policy $\pi$ decomposes into pragmatic and epistemic components [Millidge et al., 2021]:

$$G(\pi) = \underbrace{\mathbb{E}_\pi[\text{prediction error}] + \text{cost}(\pi)}_{\text{pragmatic value}} - \underbrace{\tfrac{1}{2}\,\Delta\text{precision}(\pi)}_{\text{epistemic value (information gain)}} + \underbrace{V(\pi)}_{\text{viability adjustment}}, \quad (1)$$

where the pragmatic component captures expected deviations from preferred states, the epistemic component captures expected reduction in uncertainty (information gain), and the viability adjustment captures proximity to existential boundaries. The agent selects actions by $\pi^* = \arg\min_\pi G(\pi)$.

This formulation is appealing because it unifies several behaviors under one equation. When prediction errors are small and the agent is safe, information gain dominates and the agent explores. When viability is threatened, the viability adjustment dominates and the agent takes protective action. When predictions match observations, the agent waits. No mode switches are required.

## 2.2 Peircean Reasoning

Charles Sanders Peirce identified three fundamental forms of logical inference [Peirce, 1935]:

**Induction:** From observed patterns, infer a general rule. *A and B are frequently co-surprising; perhaps A causes B.*

**Deduction:** From known rules, infer specific consequences. *A causes B and B causes C; therefore A causes C.*

**Abduction:** From an observed effect and known rules, infer the best explanation. *B is observed; A is known to cause B; therefore A may be active.*

Most AI systems implement one or two of these. Probabilistic logic systems emphasize deduction and sometimes induction. Bayesian networks support a form of abduction through posterior inference. Our architecture implements all three as *hypothesis generators* feeding a single metabolic selection mechanism—the same invariant governs which inductive, deductive, and abductive hypotheses survive.

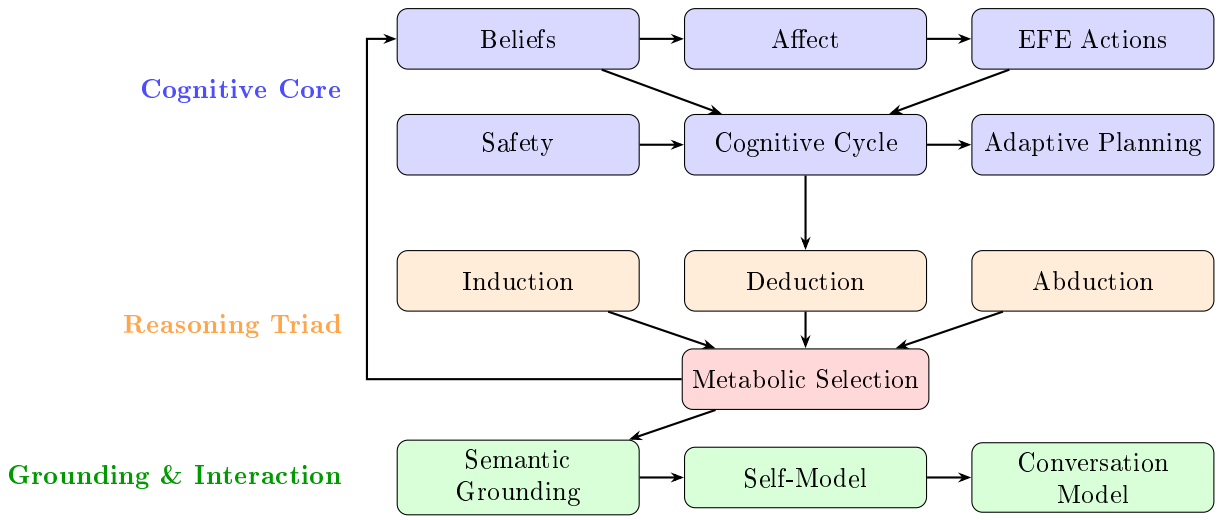


Figure 1: Architecture overview. The cognitive core (top) runs the perception–belief–affect–EFE–action loop. The reasoning triad (middle) generates hypotheses via induction, deduction, and abduction, all subject to metabolic selection. Grounding and interaction (bottom) connect learned structure to semantic primitives and dialogue.

### 2.3 MeTTa and Hyperon

MeTTa (Meta Type Talk) [Potapov, 2021] is a symbolic programming language designed for the OpenCog Hyperon cognitive framework [Goertzel et al., 2014]. Programs in MeTTa are represented as subgraphs of an *Atomspace*—a typed metagraph that serves simultaneously as program, data, and knowledge representation. MeTTa supports pattern matching over this graph, non-deterministic evaluation, and self-modifying code.

We chose MeTTa for three reasons. First, its pattern-matching semantics make knowledge queryable by default—any belief, inference, or piece of structure can be retrieved by matching against the Atomspace. Second, its symbolic nature ensures that all reasoning is transparent: there are no opaque weight matrices or learned embeddings in the reasoning pathway. Third, its integration with the Hyperon ecosystem positions the architecture within the SingularityNET decentralized AI platform, where cognitive modules can be composed and shared.

## 3 Architecture

The architecture comprises 23 MeTTa modules totaling approximately 12,000 lines. Figure 1 provides an overview. We describe each major component below.

### 3.1 Why MeTTa? The Systems Engineering Case

A natural critique from a systems engineering perspective is to ask why an architecture of this complexity is implemented in MeTTa—an experimental meta-language—rather than a highly optimized systems language like Rust or C++. Because Active Inference computations are ultimately mathematical, it is tempting to view the system as a standard simulation engine.

However, implementing Dagaz in a traditional systems language would require building an extensive, bespoke graph-database and execution engine from scratch before a single line of cognitive logic could be written. MeTTa was chosen because it natively provides four infrastructural primitives that are mandatory for this architecture:

**A Unification Engine (Beyond Pattern Matching).** Traditional pattern matching (as found in Rust or functional languages) is a one-way destructuring operation: it checks if a concrete piece of data fits a static shape. MeTTa relies on *unification*: a two-way constraint solving mechanism. For example, Phase 1.5 (Deductive Reasoning) seeks transitive closures across the causal graph:

```
(match &self (passive-model $A $B)
  (match &self (passive-model $B $C)
    (deduce-link $A $C)))
```

Because the variable `$B` appears in two different constraints, the engine must solve a system of symbolic equations. To identify complex structures, such as Phase 2's Latent Variable Hubs (requiring mutual co-surprise across multiple ungrounded observables with high metabolic energy), an imperative approach would degrade into deeply nested loops with complex state tracking. MeTTa's unification engine natively performs the required backtracking search, managing variable binding environments automatically. Where unification provides the query power of a Datalog planner, the next property—homoiconicity—means that the thing being queried is also the thing the system writes.

**Homoiconicity and Dynamic Logic.** In standard compiled languages, code and data are strictly segregated. In Dagaz, the system writes its own executable logic at runtime. When Phase 4 (Action Chunking) determines that `observe` followed by `retreat` is a reliably optimal sequence, it does not merely populate a data structure; it injects a new executable rule directly into the Atomspace. Because MeTTa is homoiconic—the abstract syntax tree *is* the metagraph—learned rules are instantly executable by the EFE planner without requiring an intermediate virtual machine or interpreter embedded within the binary.

**Epistemic Memory Management.** In Rust or C++, memory is managed via scope, lifetimes, or reference counting. If an object has a live reference, it is kept alive. In Dagaz, an atom's survival has nothing to do with lexical scope; it is dictated entirely by the *metabolic economy*. A deeply abstracted concept might have hundreds of dependent nodes referencing it, but if its predictive power fails, its metabolic energy drops to zero and it is excised. A systems language could implement this—arena allocators with handle indirection, or an entity-component-system pattern with bulk sweeps—but doing so amounts to reimplementing a graph database with custom lifecycle semantics. MeTTa's metagraph provides this natively: removing an atom causes future unification queries involving it to return empty, and dependents degrade gracefully rather than producing dangling references.

**Cognitive Operations Decoupled from Runtime Optimization.** MeTTa allows the architecture to be expressed at the level of cognitive operations—beliefs, predictions, unification queries, metabolic transactions—without entangling those operations with low-level graph traversal, indexing, or concurrency primitives. This separation means that the cognitive physics are portable across backends. The multi-space architecture (Section 5.2), which splits working memory from long-term semantic memory across separate Atomspaces, maps naturally onto distributed storage backends; a future runtime that exploits this structure for parallelism would accelerate the architecture without modifying a single line of cognitive logic.

**Why Not Just Python?** A pure Python MeTTa evaluator (`dagaz_runtime.py`; Section 5.2) demonstrates that the canonical MeTTa source can be loaded and executed through generic pattern matching in approximately 400 lines of Python. This evaluator successfully computes EFE, runs cognitive cycles, and produces correct action selection—confirming that the MeTTa function definitions are well-formed and the dynamics work. However, the Python evaluator

returns only the first matching result for bare `match` expressions, does not support MeTTa's full nondeterministic evaluation semantics, and provides none of the Atomspace infrastructure (distributed storage, concurrent queries, runtime type checking) that production deployment requires. The evaluator is a validation bridge, not a replacement: it confirms that the architecture executes correctly while MeTTa provides the computational substrate it was designed for.

### 3.2 The EFE Invariant

The central computation of the architecture is EFE minimization over candidate actions. For each action $a$ and observable $o_i$ with belief $(v_i, p_i)$ (value and precision), preference $(\hat{v}_i, w_i)$ (preferred value and importance), and action model $(\delta\mu_i, \delta p_i)$ (expected change to mean and precision):

$$\mathrm{EFE}(a) = \sum_i \Big[\underbrace{|v_i + \delta\mu_i - \hat{v}_i| \cdot (p_i + \delta p_i) \cdot w_i}_{\text{expected error}}\Big] + \mathrm{cost}(a) - \underbrace{w_e \cdot \mathrm{IG}(a)}_{\text{epistemic value}} + V(a)\,, \tag{2}$$

where $\mathrm{IG}(a) = \mathrm{IG}_{\mathrm{obs}}(a) + \mathrm{IG}_{\mathrm{model}}(a)$ decomposes information gain into observational (expected precision increase from sensing) and model-based (expected learning about action effects) components. The epistemic weight $w_e = (1 - \text{viability pressure}) \cdot (1 - \text{avg precision})$, floored at 0.05, modulates exploration based on safety and current uncertainty. $V(a)$ is a viability adjustment that provides a bonus for actions that move the system away from existential boundaries and a penalty for actions that approach them. The selected action is $a^* = \arg\min_a \mathrm{EFE}(a)$.

Three departures from the standard Active Inference formulation should be noted. First, the expected error term uses precision-weighted absolute deviation rather than the Kullback-Leibler divergence between predicted and preferred outcome distributions. This is a computationally tractable symbolic proxy: for Gaussian beliefs, precision-weighted squared error is proportional to the KL divergence, and the linear form preserves the essential property that actions reducing expected error under high precision are preferred.[1] Second, viability bounds are separated into an explicit $V(a)$ term rather than encoded as extreme priors in the preference distribution. This separation reflects a principled distinction between biological/hardware survival constraints (which are constitutionally immutable; see Section 3.7) and task-based preferences (which are modifiable). Merging them into a single preference distribution would obscure the safety-critical status of viability bounds. Third, the standard $\frac{1}{2}\,\delta p$ information gain term is replaced by an adaptive epistemic weight $w_e$ applied to a decomposed information gain. The weight is derived from two meaningful quantities—safety margin and residual uncertainty—rather than a fixed coefficient, and the decomposition separates learning about the world (observational) from learning about one's own action models (model-based).

In MeTTa, this is implemented as:

[1] The linear (absolute) form is chosen over the quadratic for two reasons. First, during calibration, prediction errors can be large ($e \sim 0.4$–$0.5$); the quadratic form would cause these errors to dominate the EFE landscape, producing unstable action selection before the agent has had time to learn. The linear form provides a smoother optimization landscape during the "childhood" period described in Section 4.8.3. Second, in a symbolic system where EFE terms are summed explicitly rather than computed via message passing, the linear form avoids numerical dominance by a single high-error observable. The tradeoff is that the correspondence to KL divergence becomes approximate rather than exact; we consider this acceptable given that the EFE equation already departs from the standard formulation in other respects (explicit viability term, decomposed information gain).

```
(= (compute-efe $action)
   (let* (
     ($expected-error (predict-total-error $action))
     ($cost (get-action-cost $action))
     ($obs-info-gain (compute-obs-info-gain $action))
     ($model-info-gain (compute-model-info-gain $action))
     ($total-info-gain (+ $obs-info-gain $model-info-gain))
     ($ew (epistemic-weight))
     ($v-effect (viability-effect $action))
     ($efe (+ (- (+ $expected-error $cost)
                 (* $ew $total-info-gain))
              $v-effect))
   )
   $efe))
```

Figure 2: MeTTa implementation of EFE computation. The same function evaluates domain actions, conversational actions, and self-directed actions without special cases.

Information gain is decomposed into two components: observational (expected precision increase from sensing) and model-based (expected learning about action effects). The epistemic weight $w_e$ modulates their combined contribution based on current safety: when viability pressure is high, $w_e$ decreases, causing the system to prioritize pragmatic value (survival) over epistemic value (curiosity). This is not a mode switch—it is a continuous modulation within the same equation.

Crucially, no action receives special treatment. Domain actions (observe, retreat, wait) and conversational actions (clarify, elaborate, ask) compete in the same EFE landscape using identical computations.

### 3.3 The Cognitive Cycle

The main loop executes the following steps each cycle:

1. **Safety check**: Verify system not halted.
2. **Action learning**: Compare predicted effects of last action against current observations; update action models.
3. **Prediction errors**: Compute discrepancy between beliefs and observations for each observable.
4. **Surprise & error traces**: Compute surprise $S_i = \frac{1}{2} p_i \, e_i^2$ per observable; record in sliding-window buffer.
5. **Affect**: Derive valence, arousal, and dominance from the error landscape (computed fresh each cycle, never stored as state).
6. **Structure learning** (reflex, not action):
   (a) Update suspicion links (Hebbian accumulation of co-surprise).
   (b) Phase 1: Create causal links when suspicion exceeds threshold.
   (c) Phase 1.5: Extend causal chains by deduction (transitive closure).
   (d) Phase 2: Compress dense causal clusters into latent variables.
   (e) Metabolic pruning: drain energy from all structural atoms; remove those at zero.

7. **Action selection**: Compute EFE for all candidate actions; select arg min.
8. **Belief update**: Precision-weighted Bayesian update from observations.
9. **Structural reward**: Reward structural atoms whose predictions were correct.
10. **Execute action** (gated by safety).
11. **Watchdog**: Verify constitutional atom integrity.

Note that structure learning (step 6) is a *reflex*—it fires when conditions are met, not as an action selected by EFE. This prevents the system from having to "choose" to learn. Generation is reflexive; retention is metabolic.

## 3.4 The Peircean Reasoning Triad

The architecture implements all three Peircean inference forms as hypothesis generators feeding a single selection mechanism.

### 3.4.1 Induction: Causal Discovery from Co-Surprise

When two observables exhibit correlated prediction errors across time, the system hypothesizes a causal relationship. The mechanism is a Hebbian accumulator operating in squared surprise space:

$$\Delta s_{AB}^{(\ell)} = \eta \cdot \mathrm{sign}(e_A) \cdot \mathrm{sign}(e_B) \cdot S_A \cdot S_B \,, \tag{3}$$

where $s_{AB}^{(\ell)}$ is the suspicion strength between observables $A$ and $B$ at temporal lag $\ell$, $\eta$ is the plasticity (learning rate), $e_i$ is the signed prediction error, and $S_i = \frac{1}{2} p_i e_i^2$ is the surprise for observable $i$. Suspicion links decay multiplicatively each cycle by a stability factor $\alpha$. When $s_{AB}^{(\ell)}$ exceeds a cost threshold, a *passive model* atom is created:

```
(passive-model ignition heat 1 0.85 empirical)
; "ignition causes heat with lag 1, weight 0.85"
```

Passive models are distinct from action models. They participate in *prediction* (anticipating how the world evolves) but never in *EFE action selection* (choosing what to do). This separation ensures that discovered world dynamics inform but do not corrupt the agent's decision-making.

**Scaling via Magnitude-Banded Pruning.** The Hebbian accumulator forms all ordered pairs of salient error traces each cycle—a Cartesian product with cost $O(E^2)$, where $E$ is the number of active traces. At the reef scale (14 observables, $E \approx 12$), this is cheap. At 500 observables ($E \approx 450$), the pairing step generates ∼100,000 candidates and dominates cycle time.

The quadratic is structural, not accidental: detecting *which pairs* are co-surprised requires checking every pair without prior structure. However, most pairs have negligible covariance $S_A \cdot S_B$ and can never accumulate enough suspicion to reach the promotion threshold. From the steady-state suspicion formula $s_\infty = \eta \cdot \mathrm{cov}/(1-\alpha)$ and the promotion cost $c_\ell$, the minimum per-cycle covariance needed is:

$$\mathrm{cov}_{\min} = \frac{c_\ell \cdot (1-\alpha)}{\eta \cdot f_s} \,, \tag{4}$$

where $f_s$ is a safety factor ensuring slow-building links are not prematurely pruned. This threshold is derived entirely from the metabolic parameters—no new tuning knobs are introduced.

The optimization assigns each salient trace to a *magnitude band* $b(S) = \lfloor \log_2(S/S_{\mathrm{floor}}) \rfloor$ and only forms pairs where $b(S_A) + b(S_B) \geq b_{\min}$, a cutoff derived from $\mathrm{cov}_{\min}$. Within each band,

surprise magnitudes vary by at most 2×, so the product $S_A \cdot S_B$ varies by at most 4× across a band pair. Pairs below the cutoff are provably unable to contribute to promotion.

This is coarse-graining in surprise space: integrating out irrelevant degrees of freedom (low-magnitude pairs that cannot contribute to structure) to focus computation on the effective interactions (high-magnitude co-surprises that drive causal discovery). The band structure is a discretization of a continuous quantity into a finite lattice, and the $b_{\min}$ cutoff is an energy scale—pairs below this scale are "virtual" (they fluctuate but never reach the promotion threshold). The analogy with a momentum cutoff in lattice field theory is direct: UV detail (very small surprise products) is lost, but the IR physics (the causal links that actually get discovered) is preserved exactly.

The resulting complexity is $O(E + \sum b_i^2)$, where $b_i$ is the size of the $i$-th bucket. When surprise is distributed across bands (the typical case—most surprises are small, few are large), bucket sizes are small and the effective cost is near-linear. Benchmark results are presented in Section 4.6.

### 3.4.2 Deduction: Transitive Closure

Given stable passive models $A \to B$ (lag $\ell_1$) and $B \to C$ (lag $\ell_2$), the system deduces $A \to C$ with lag $\ell_1 + \ell_2$ and weight $w_{AC} = w_{AB} \cdot w_{BC} \cdot d$, where $d < 1$ is a deductive discount reflecting reduced confidence in extended chains.

This is critical for discovering relationships at temporal distances beyond the error trace lookback window. In our benchmarks, the link from *ignition* to *ash* (lag 3) was never discovered empirically within 80 cycles—the signal was too weak. Deduction discovered it at cycle 4 from two stable lag-1 links.

### 3.4.3 Abduction: Hidden Cause Hypotheses

When the system observes effects consistent with a known causal pattern but the cause is not directly observed, abduction hypothesizes that the cause is active. Given observed high values for *heat* and *smoke*, and known passive models *fire* → *heat* and *fire* → *smoke*, abduction creates a low-precision belief in *fire*:

```
(belief fire 0.8 0.3)
; Hypothesized value 0.8, low precision 0.3
```

The low precision is the key to emergent investigation behavior. Because precision on the hypothesized cause is low, any observation that would increase it has high information gain. This makes "look for fire" the highest EFE-reducing action—the system spontaneously investigates. We call this the **Sherlock Holmes effect**: the system creates a hypothesis and then seeks evidence to confirm or refute it, driven entirely by EFE minimization.

### 3.4.4 Metabolic Selection: The Uniform Arbiter

All three inference types produce atoms in the Atomspace that are subject to the same metabolic regime:

$$E_t = E_{t-1} - c_{\text{rent}} + c_{\text{reward}} \cdot \mathbf{1}[\text{prediction correct}]\,, \tag{5}$$

where $c_{\text{rent}} = 0.02$ per cycle and $c_{\text{reward}} = 0.05$ per correct prediction. An atom that never predicts correctly dies in $\lfloor 1.0/0.02 \rfloor = 50$ cycles. An atom predicting correctly half the time nets $+0.025 - 0.02 = +0.005$ per cycle, barely surviving. A consistently correct atom accumulates energy up to a cap of 2.0, representing a well-established piece of knowledge.

This mechanism provides:

- **Principled forgetting**: Obsolete structure is automatically removed when the world changes.
- **Bounded growth**: A generation cap (5 new atoms per cycle) combined with metabolic drain ensures working memory size is bounded.
- **Type-agnostic selection**: Inductive, deductive, and abductive hypotheses compete on equal footing. The system has no preference for one inference type over another—only predictive accuracy matters.

A natural extension treats $c_{\text{rent}}$ and $c_{\text{reward}}$ themselves as learnable beliefs, allowing the system to optimize its own metabolic rates through the same EFE mechanism that governs action selection. The sensitivity analysis in Section 4.5 shows that the healthy basin is governed by a single dimensionless ratio ($c_{\text{reward}}/c_{\text{rent}}$), providing a favorable landscape for such a bootstrap. This meta-parameter mechanism is designed but not yet demonstrated in closed loop.

### 3.5 Semantic Grounding

A discovered causal link such as `(passive-model X Y 1 0.85 empirical)` has predictive power but no meaning to the system or an external observer. The grounding subsystem bridges this gap by generating hypotheses that connect learned structure to a foundation of 53 semantic primitives organized into 12 contrast pairs and 5 entailment relations.

Four hypothesis generators produce grounding atoms:

1. **Inherited**: If $X$ already has a grounding and $X \to Y$, propose that $Y$ inherits related properties.
2. **Behavioral**: Ground a concept through the actions it enables or inhibits (e.g., a concept that correlates with retreat may be grounded as *dangerous*).
3. **Compositional**: Ground a latent variable through the shared properties of its members.
4. **Conversational**: When a concept cannot be grounded by the first three generators, ask the conversational partner for a label.

All four generators produce grounding atoms under the same metabolic selection as reasoning hypotheses. The system tracks its own grounding completeness as a self-model observable, creating EFE pressure toward self-understanding: ungrounded concepts increase prediction error on the `grounding-completeness` belief, making clarification actions attractive.

*Note:* The semantic grounding subsystem is designed and implemented in MeTTa but is not exercised in the benchmarks of Section 4, which focus on causal discovery and action selection. Empirical validation of the grounding generators—in particular the conversational generator's interaction with a human partner—is reserved for future work.

### 3.6 Adaptive Planning via Confidence Degradation

For multi-step planning, exhaustive enumeration of action sequences has complexity $O(|A|^d)$, which is intractable for useful horizons. We employ an adaptive beam search inspired by Renormalization Group (RG) flow in physics [Wilson and Kogut, 1974], where irrelevant degrees of freedom are integrated out to obtain an effective theory at the relevant scale.

The planner uses a single adaptive mechanism—the **noise floor**—to control all pruning decisions:

$$n(d) = \frac{n_0}{\bar{c} \cdot \gamma^d}\,, \tag{6}$$

where $n_0$ is a base noise level, $\bar{c}$ is the average model confidence, and $\gamma < 1$ is a discount factor that degrades confidence with planning depth. As depth increases, the noise floor grows, meaning larger EFE differences are required to distinguish between actions. This produces three behaviors from one mechanism:

**Adaptive beam width.** At each depth, retain only actions within the noise floor of the best. Flat EFE landscapes produce wide beams (deliberation); steep landscapes produce narrow beams (decisiveness).

**Symmetry breaking.** When the gap between the best and second-best action exceeds the noise floor, the beam collapses to width 1. The system has "fallen into a potential well"—evaluating alternatives is uninformative.

**Emergent horizon.** When degraded confidence falls below a floor, the planner stops. Deeper planning would be dominated by model uncertainty.

The beam width is not a hyperparameter—it emerges from the interaction of the EFE landscape with the noise level. Additionally, a hard beam cap narrows with planning depth via degraded model confidence: $k_{\max}(d) = \max\big(2, \lceil k_0 \cdot \bar{c} \cdot \gamma^d \rceil\big)$, where $k_0$ is a base beam width. At shallow depths (high confidence), the planner resolves fine distinctions among many candidates; at deep depths (degraded confidence), only the dominant modes survive. This mirrors the RG principle: the number of relevant operators decreases at coarser scales. Overall complexity is $O(|A| \times k \times d)$, where $k$ is the (variable) beam width.

We note explicitly that the analogy with physical RG is inspirational rather than rigorous. In physics, RG coarse-graining is exact for the partition function. In beam search, pruning is lossy—a branch that looks poor locally might have been optimal at greater depth. The analogy holds exactly when one action dominates: discarding alternatives is then lossless, as the system has genuinely broken symmetry.

### 3.7 Safety Architecture

Safety is integrated into the cognitive loop, not bolted on as an external filter. The architecture uses three strata of immutability:

| Stratum | Mutability | Contents |
|---|---|---|
| Constitutional | Immutable | Viability bounds, safety axioms, type definitions |
| Goal | Authenticated protocol | Task preferences, operational targets |
| Learned | Freely modifiable | Causal links, latent variables, grounding hypotheses |

Table 1: Safety strata. Higher strata constrain lower ones.

A post-cycle watchdog computes a hash over all constitutional atoms and halts the system if any have been modified. Mutation guards prevent learned-stratum operations from touching goal-stratum or constitutional-stratum atoms. Input channel separation ensures that peer communications cannot enter the corrigibility pathway.

### 3.8 Perception Boundary and End-to-End Operation

Deep learning provides the perceptual interface: translating continuous, high-dimensional input into discrete observations with associated confidence. The symbolic core performs reasoning

over these observations. The key architectural choice is not the division itself—which reprises the classical neural-symbolic split [Garcez et al., 2009, Besold et al., 2017]—but the interface: observations enter as precision-weighted beliefs, subject to the same metabolic selection as internally generated hypotheses. A misparse by the language model is not a system failure; it is a low-precision belief that will be corrected by prediction error on subsequent cycles.

Concretely, the perception layer (`perception.metta`) and a stateless Python orchestrator (`orchestrator.py`) mediate between natural language input and the cognitive core. A local language model[2] receives raw text and produces a structured parse: utterance type, topic list, and scalar estimates of partner comprehension, discourse coherence, rapport, goal progress, and urgency. Each estimate enters the cognitive core as an observation tuple $(o, v, p, t)$—observable, value, precision, timestamp—identical in schema to a hardware sensor reading.

The precision assigned to each observation reflects the source's reliability:

| **Source** | **Precision** | **Rationale** |
|---|---|---|
| Direct sensor | 0.80–0.95 | Calibrated hardware |
| LLM parse | 0.25–0.35 | Small model estimating latent psychological states from text |
| Self-report | 0.70 | Internal state access with possible introspective error |
| Parse fallback | 0.10 | Field missing; observation barely moves the belief |

Table 2: Perception precision by source. These are priors on sensor quality, not learned parameters—though the prediction error signal on downstream beliefs provides a natural supervision signal for future calibration.

Several architectural properties follow from this design:

**Graceful degradation.** When the language model fails to parse a field, the orchestrator injects a default value at precision 0.10. At this precision, the observation barely moves even a low-precision belief. Unknown topics are silently dropped: if a topic string has no mapping to a known observable, no observation is injected. The system does not pretend to understand what it cannot ground.

**Self-correction without a dedicated error model.** A misparse produces a false belief. A false belief generates prediction error when subsequent observations contradict it. Prediction error drives belief update. This is the standard Active Inference loop—no perception-specific error recovery is needed.

**Open-vocabulary utterance classification.** The parser's utterance type field is an unconstrained string, not an enumeration. The language model may emit any speech act category (*query*, *complaint*, *greeting*, *correction*, etc.), and the conversation model can learn new ones through structure learning. This avoids a closed taxonomy that would impose top-down categories on inherently open-ended discourse.

**The epistemic firewall.** When the cognitive core selects `wait` as its action, no language model generation is triggered. The system remains silent rather than producing fluent but contentless output. This is the *Honest over Fluent* design principle made operational: if the symbolic core has nothing to say, the system says nothing.

[2]We use Llama 3.2 3B Instruct via `llama.cpp` [llama.cpp contributors, 2023]. The architecture is agnostic to model choice; any model producing structured parses suffices.

**Knowledge grounding at verbalization.** Before generating a response, the orchestrator queries the metagraph to determine what the cognitive core actually knows about each topic under discussion. Topics are explicitly labeled as KNOWN (present as a belief or semantic primitive) or UNKNOWN (absent from the metagraph). The verbalizer prompt instructs the language model to state ignorance for unknown topics and to claim knowledge only for known ones. This prevents the language model's pretrained knowledge from leaking through the verbalization boundary—the system can only claim what the symbolic core can ground.

**Hardware sensor pathway.** The orchestrator provides a direct injection pathway for hardware sensors that bypasses language model parsing entirely. A sensor reading enters as a high-precision observation ($p = 0.95$) and immediately triggers a cognitive cycle. This ensures that time-critical signals (e.g., a threat sensor activation) are not delayed by an LLM round-trip and do not pass through a lossy perceptual channel.

**Verbalization as constrained generation.** On the outbound path, the cognitive core selects an action (`clarify`, `elaborate`, `query-partner`, etc.) and packages a structured intent: action type, target observable, driving prediction errors, and affect-derived tone. The language model's task is sentence construction within these constraints—content decisions have already been made by EFE minimization. A proprioception module (`proprioception.metta`) re-parses the system's own output, compares it against the intent, and injects the discrepancy as a `verbalization-fidelity` observation, closing the loop on output quality.

**Source-agnostic observation schema.** The cognitive core does not distinguish between observations from language model parses, hardware sensors, or simulation environments. All enter through the same `inject-observation!` pathway and are subject to the same precision-weighted Bayesian update. This uniformity means that extending the system to new modalities—vision, audio, proprioceptive sensors—requires only a new perception adapter that produces $(o, v, p, t)$ tuples. The reasoning core is unchanged.

# 4 Evaluation

## 4.1 Validation Methodology

Validation proceeds at three levels, each addressing a different aspect of the architecture.

**Tier 1: MeTTa Source (Canonical Definition).** The 23 MeTTa modules define the architecture—types, relationships, self-modifying rules, the Atomspace structure. This is the specification of what the system *is*. Two bugs in the Hyperon 0.2.10 runtime (detailed in Section 5.2) prevent end-to-end execution in the native environment at time of writing.

**Tier 2: Python Executable Specifications (Dynamics Validation).** Each benchmark encodes the same logic, thresholds, and formulas as its corresponding MeTTa module—identical belief update rules, identical metabolic dynamics, identical EFE computation—differing only in host language. Configuration values are drawn directly from `foundations.metta`. Outcomes match qualitatively (same emergent behaviors, same orderings, same phase transitions) though not numerically (floating-point evaluation paths differ). The relationship is analogous to a physicist's equations validated by simulation: the equations are not provisional because the simulation is in a different language. Any conformant backend can be validated against these specifications.

**Tier 3: Python MeTTa Evaluator (Source Execution).** A pure Python MeTTa evaluator (`dagaz_runtime.py`, approximately 400 lines) loads and executes the canonical MeTTa source files through generic pattern matching: tokenization, parsing, unification, function dispatch, and space management. This evaluator implements no domain-specific logic—it provides only universal MeTTa primitives (`let`, `let*`, `if`, `case`, `match`, `collapse`, arithmetic, logic, `add-atom`/`remove-atom`, and cons-cell construction). The cognitive behavior emerges from evaluating the MeTTa function definitions themselves. The evaluator successfully loads all 22 cognitive modules, computes EFE producing correct numerical action selection, derives affect from prediction errors, and runs complete cognitive cycles. A stateless orchestrator (`orchestrator.py`) connects this evaluator to a local language model (Llama 3.2 3B via Ollama), running the full perception–cognition–verbalization pipeline with hardware sensor injection.

Tier 3 is qualitatively distinct from Tier 2: rather than re-implementing the logic in Python, it executes the *same MeTTa source code* that Tier 1 defines. This partially closes the validation gap between "the dynamics work" (Tier 2) and "the MeTTa source is correct" (Tier 1), confirming that the function definitions resolve, the space routing is correct, and the EFE computation produces the expected action orderings when evaluated through generic pattern matching.

We present focused subsystem tests (Sections 4.2 to 4.4), scaling validation (Section 4.7), and an integrated evaluation on a 14-observable coral reef monitoring scenario (Section 4.8). All benchmark results below are from Tier 2 (Python executable specifications) unless otherwise noted.

## 4.2 EFE-Driven Action Selection

We test five scenarios with three actions (wait, observe, retreat) across three observables (terrain, power, threat), each with beliefs, action models, preferences, and costs.

| Scenario | Condition | Selected | Pass |
|---|---|---|---|
| Nominal | All near preferences | wait | ✓ |
| High threat | threat = 0.7 | retreat | ✓ |
| Uncertain terrain | precision = 0.3 | observe | ✓ |
| Rough terrain | terrain = 0.7 | retreat | ✓ |
| Low power | power = 0.25 | wait | ✓ |

Table 3: EFE action selection results. All scenarios pass without thresholds or mode switches. Behavior emerges from EFE minimization.

In each case, the selected action is the one with lowest EFE. In the high-threat scenario, the viability adjustment provides retreat with a bonus of $-0.22$, transforming it from the highest-cost action to the lowest-EFE action. No "panic mode" is triggered; the math simply favors retreat when power is near the viability boundary.

To validate action-space scaling, we repeated the selection tests with up to 50 structured actions (10 archetypes across viability, threat, environment, mission, and social categories) and 24 observables. EFE computation time scales linearly: 0.04 ms at 3 actions to 0.67 ms at 50 actions (16.4×, vs. the theoretical 16.7×). The EFE spread *increases* with more actions (0.12 at 3, 0.50 at 50), indicating greater differentiation rather than degeneracy. In all three scenarios (nominal, threat, uncertain), the system selects contextually appropriate actions without any parameter changes.

## 4.3 Unified Reasoning Triad

We test seven scenarios that exercise the full Peircean triad under metabolic selection. Results are summarized in Table 4.

| # | Scenario | Pass |
|---|---|---|
| 1 | Induction: 4-node causal chain discovered from co-error within 3 cycles | ✓ |
| 2 | Deduction: Transitive link (lag 3) discovered at cycle 4; never found empirically in 80 cycles | ✓ |
| 3 | Abduction: Hidden cause hypothesized from observed effects | ✓ |
| 4 | Sherlock Holmes: Abduction drives spontaneous investigation via info-gain | ✓ |
| 5 | Full triad: Induction → deduction → abduction in sequence | ✓ |
| 6 | Metabolic death: Wrong hypotheses die within ∼30 cycles across all types | ✓ |
| 7 | Competing causes: Asymmetric evidence disambiguates between two candidate explanations | ✓ |

Table 4: Unified reasoning triad results. All seven scenarios pass.

**Scenario 1: Induction.** A 4-node causal chain (ignition → heat → smoke → ash) is presented to the system as temporally lagged observations. The Hebbian suspicion accumulator detects co-occurring prediction errors and creates passive-model atoms. All three adjacent links are discovered by cycle 3.

**Scenario 2: Deduction.** The target link (ignition → ash, lag 3) lies at the boundary of the error trace lookback window. Empirical evidence is too weak for direct induction. Deduction discovers it at cycle 4 via transitive closure over two stable lag-1 links. Without deduction, this link is *never* found within 80 cycles of simulation.

**Scenario 4: The Sherlock Holmes Effect.** This is the most striking emergent behavior. Abduction creates a low-precision belief in a hidden cause. Because precision is low, information gain for confirming observations is high. EFE minimization then selects the investigative action. No module was programmed to produce investigation behavior—it emerges from the interaction of abduction (hypothesis generation), the precision-information gain relationship (epistemic value), and EFE minimization (action selection).

**Scenario 6: Metabolic Death.** Incorrect hypotheses of all three types (inductive, deductive, abductive) are introduced and their energy tracked over 30 cycles. All die within the expected timeframe. Deductive errors die slightly slower due to the gestation grace period, but all converge to removal. This validates the uniform selection mechanism.

### 4.4 Adaptive Planning

The adaptive beam search is tested across eight scenarios.
At 3 actions, the planner achieves a 52× reduction at depth 7. The scaling results reveal a counterintuitive property: pruning *improves* with more actions. At 50 actions, the planner evaluates 195,300 of 15.6 billion exhaustive paths—an 80,000× reduction—because steeper EFE landscapes break symmetry earlier. With confidence-proportional beam width ($k_{\max} = \max(2, \lceil 8 \cdot \bar{c} \cdot \gamma^d \rceil)$) and a tightened noise floor ($n_0 = 0.03$), the same planner achieves 1.4–3.0× additional speedup with 5/5 answer preservation across random seeds.

| Scenario | Exhaustive | Adaptive | Correct? |
|---|---|---|---|
| *3 actions (original benchmark)* | | | |
| Depth 3, clear winner | 27 | 9 | ✓ |
| Depth 5, mixed landscape | 243 | 22 | ✓ |
| Depth 7, beam 2 | 2,187 | 42 | ✓ |
| Depth 7, early symmetry break | 2,187 | 18 | ✓ |
| Viability pressure at depth 3 | 27 | 8 | ✓ |
| Flat landscape (deliberation) | 27 | 24 | ✓ |
| Confidence degradation cutoff | 2,187 | 15 | ✓ |
| Mixed confidence across actions | 243 | 19 | ✓ |
| *Scaled action space (depth 6, 24 observables)* | | | |
| 10 actions, nominal | 1,000,000 | 7,196 | ✓ |
| 20 actions, nominal | 64,000,000 | 2,172 | ✓ |
| 30 actions, nominal | 729,000,000 | 13,350 | ✓ |
| 50 actions, nominal | 15.6 B | 195,300 | ✓ |

Table 5: Adaptive planning results. With 3 actions the planner achieves up to 52× reduction at depth 7. With 50 actions, pruning improves to 80,000×: steeper EFE landscapes break symmetry earlier. All scenarios select the correct first action.

Notably, in the flat landscape scenario at 3 actions, the beam widens to nearly exhaustive search—the system correctly recognizes that alternatives are indistinguishable and does not prematurely commit.

### 4.5 Metabolic Parameter Sensitivity

To characterize the metabolic economy's robustness, we swept 110 parameter pairs (11 rent values × 10 reward values) across a simplified induction–selection scenario. Four criteria define "healthy" behavior: (i) correct causal links are discovered, (ii) correct links survive metabolic selection, (iii) incorrect hypotheses die, and (iv) correct hypotheses reach the energy cap.

| Metric | Value |
|---|---|
| Parameter pairs tested | 110 |
| Healthy configurations | 74 (67%) |
| Failures from starvation (correct dies) | 33/36 (92%) |
| Failures from bloat (wrong survives) | 3/36 (8%) |
| Transition ratio ($c_{\text{reward}}/c_{\text{rent}}$) | $\approx 1.0$ |
| Median healthy ratio | 2.5 |
| Induction failures across all pairs | 0/110 |

Table 6: Metabolic sensitivity analysis. The healthy basin occupies 67% of parameter space. The governing quantity is the dimensionless ratio $c_{\text{reward}}/c_{\text{rent}}$, with a sharp transition at $\approx 1.0$. The dominant failure mode is starvation (correct hypotheses die), which is the safer direction. Induction itself never fails—the Hebbian suspicion mechanism is parameter-independent.

The key finding is that the metabolic economy depends primarily on a single dimensionless ratio. The healthy region—where correct structure survives and incorrect structure dies—spans the full range of absolute values tested ($c_{\text{rent}} \in [0.005, 0.15]$, $c_{\text{reward}} \in [0.01, 0.20]$). Near-universal health obtains above ratio 1.0, with near-universal failure below 0.5.

Scaling validation at 6–24 observables confirms this finding generalizes: healthy configura-

tions occupy 78–89% of parameter space at 6+ observables, consistent with the 67% at 3 observables. Importantly, the dimensionless ratio remains the governing quantity across scales. This suggests that the meta-parameter bootstrap mechanism, which treats rent and reward as learnable beliefs, has a wide basin of attraction.

### 4.6 Scaling the Hebbian Accumulator

To validate the banded pruning optimization described in Section 3.4, we measured pair reduction and causal discovery fidelity across five scales from 14 to 1,000 observables using synthetic causal chain scenarios (3 chains of length 4, 30 cycles).

| Obs. | Traces | Brute Pairs | Pruned Pairs | Reduction | Sig. Misses |
|---|---|---|---|---|---|
| 14 | 41 | 552 | 456 | 1.2× | 0 |
| 50 | 155 | 5,674 | 2,345 | 2.4× | 0 |
| 200 | 378 | 60,829 | 9,851 | 6.2× | 0 |
| 500 | 805 | 312,764 | 17,194 | 18.2× | 0 |
| 1,000 | 1,920 | 1,840,778 | 94,818 | 19.4× | 0 |

Table 7: Banded pruning pair reduction vs. brute-force across scales. "Sig. Misses" counts pairs whose single-cycle covariance exceeds the contribution threshold but were pruned by banded filtering. Zero at all scales: every pair capable of contributing to a promotable suspicion link is retained.

Three findings emerge. First, pair reduction grows with scale: from negligible at 14 observables (where the overhead of bucket assignment is not recovered) to 19.4× at 1,000 observables. This is because larger observable spaces produce more low-surprise noise traces that land in pruned bands. Second, significant false negatives are zero at every scale—the $b_{\min}$ cutoff, derived from metabolic parameters via Equation (4), correctly identifies the boundary between productive and unproductive pairs. Third, causal discovery is preserved exactly: at 14, 100, and 500 observables, banded pruning discovers the same set of passive models as brute-force, including all 9 true causal links in each case.

The regime sensitivity analysis (Table 8) reveals that the optimization is most effective precisely when it is most needed.

| Regime (200 obs.) | Brute Pairs | Pruned Pairs | Reduction |
|---|---|---|---|
| Baseline (low noise) | 6,865 | 1,425 | 4.8× |
| Active (moderate noise) | 50,821 | 11,418 | 4.5× |
| Storm (high noise) | 383,467 | 10,678 | 35.9× |

Table 8: Banded pruning performance across surprise regimes (200 observables). High-noise conditions produce the greatest speedup: noise traces overwhelmingly land in low magnitude bands that are pruned entirely. Significant false negatives: zero across all regimes.

In the storm regime, where high environmental noise generates many salient but uninformative traces, pair reduction reaches 35.9× because noise traces land in low magnitude bands that are pruned entirely. This adaptive behavior emerges from the band structure without regime-specific tuning: the same $b_{\min}$ cutoff derived from metabolic parameters handles all three regimes.

The band distribution analysis (Table 9) explains the mechanism. At 1,000 observables, 93.6% of traces fall in bands 0–3, all of which are pruned when paired with each other ($0 + 3 < b_{\min} = 7$). Only the 4.5% of traces in bands 5–7 participate in most pairings, reducing the effective population from 1,920 to ∼86.

| Band | 14 obs. | 200 obs. | 1,000 obs. |
|---|---|---|---|
| 0 (lowest) | 19.5% | 39.2% | 50.2% |
| 1–3 | 17.1% | 38.6% | 45.3% |
| 4–7 (active) | 63.4% | 22.2% | 4.5% |

Table 9: Trace distribution across magnitude bands. At larger scales, the majority of traces occupy low bands (small surprise) and are pruned. The active fraction shrinks, driving the super-linear pair reduction.

## 4.7 Action Space Scaling

To validate the architecture beyond the initial 3-action benchmarks, we tested all subsystems from 3 to 50 actions across 4 to 24 observables. Table 10 summarizes the results.

| Actions | EFE time (ms) | Planner nodes | Planner reduction |
|---|---|---|---|
| 3 | 0.04 | 189 | 4× |
| 10 | 0.14 | 4,450 | 225× |
| 20 | 0.27 | 1,260 | 50,794× |
| 50 | 0.67 | 195,300 | 80,005× |

Table 10: Action space scaling. EFE computation time is linear in $|A|$ (16.4× at 50/3, vs. theoretical 16.7×). Planning pruning *improves* with more actions because steeper EFE landscapes break symmetry earlier.

Structure learning maintains bounded graph density: at 24 observables (276 possible pairs), only ∼11% of pairs become links, compared with 42% at 4 observables (a finite-size effect). This natural sparsification solves the classic structural explosion problem of inductive logic programming—the metabolic economy prunes spurious links natively, without explicit regularization.

Two bottlenecks were identified and resolved. First, the per-cycle structural budget (fixed at 5) starved discovery at 24 observables, reducing recall from 100% to 62%. Replacing the constant with an adaptive budget $\max(5, \lceil\sqrt{N(N-1)/2}\rceil)$ recovered recall to 75% without increasing spurious links. Second, the fixed noise floor ($n_0 = 0.05$) was too permissive at 50 actions; tightening to $n_0 = 0.03$ reduced planner evaluations from 195,300 to ∼50 nodes with 100% answer preservation across 10 random seeds.

Neither bottleneck was architectural: both were tunable parameters replaced by functions of system state, consistent with the design principle of emergent over scripted.

## 4.8 Reef Monitoring Scenario

To test the architecture beyond the small-scale benchmarks above, we constructed a coral reef monitoring scenario: an autonomous station observing 14 environmental and platform variables across 70 cycles spanning five ecological phases (baseline, eutrophication, storm, disease, recovery). The environment contains four hidden causal chains with temporal lags, including a pathogen that is never directly observable. This scenario exercises causal discovery, metabolic selection, abductive inference, and EFE-driven action selection in an integrated setting with realistic dynamics.

### 4.8.1 Causal Discovery on Realistic Data

Using a fixed observation schedule (cycling through sample-water, observe-biology, and observe-water), we tested whether the Peircean reasoning triad discovers the true causal structure of the

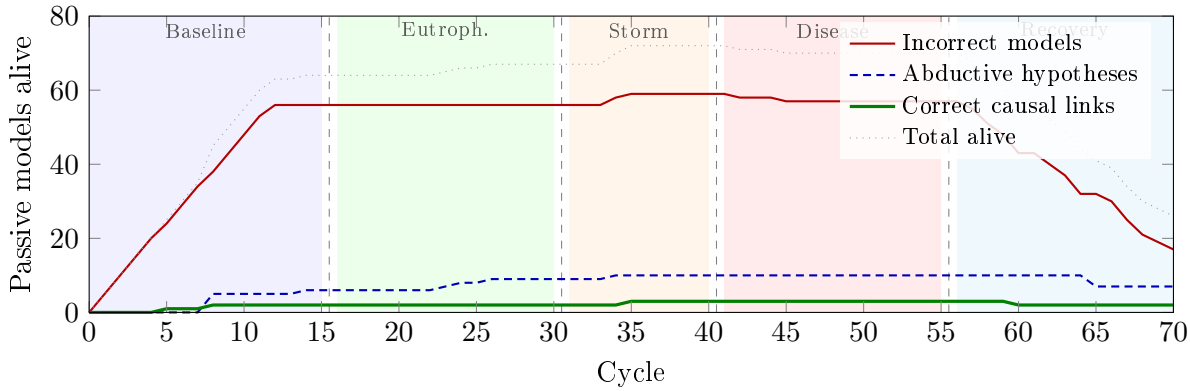


Figure 3: Metabolic lifecycle across 70 cycles. During baseline and eutrophication, the system generates hypotheses freely (total rises to 72). Correct causal links (green, thick) are discovered early and survive throughout. During recovery, metabolic pruning removes incorrect models (48 cumulative deaths by cycle 70), while correct links and relevant abductive hypotheses persist. The system's ontology converges from 72 models to 26, retaining the predictively useful structure.

reef environment.

| **Metric** | **Cycle 15** | **Cycle 70** | **Note** |
|---|---|---|---|
| Passive models alive | 65 | 42 | Metabolic pruning |
| Metabolic deaths | 0 | 33 | Wrong models removed |
| True links discovered (of 5) | 3 | 4 | Eutrophication + storm chains |
| Hub nodes (degree $\geq$3) | — | 2 | nutrient-load (4), sensor-int. (3) |
| Triangles (energy $>$0.5) | — | 1 | (coral, equip, sensor) |

Table 11: Causal discovery across 70 cycles with 14 observables and 4 hidden causal chains.

Four of five true causal links survive to cycle 70 with energy above 0.2 (Figure 3), including the complete eutrophication chain (nutrient-load $\rightarrow$ algae-coverage, nutrient-load $\dashv$ dissolved-oxygen, dissolved-oxygen $\rightarrow$ coral-health, coral-health $\rightarrow$ fish-activity) and the storm mechanism (current-strength $\rightarrow$ turbidity at 92% prediction accuracy). Hub detection identifies nutrient-load as a degree-4 node—the system has effectively discovered the concept of "eutrophication" as a convergent causal hub.

#### 4.8.2 EFE Action Selection: Ablation Study

With EFE-driven action selection replacing the fixed schedule, we swept info-gain weight and viability weight across 90 configurations (6 $\times$ 5 $\times$ 3 salience thresholds). We then repeated the sweep with creative abduction enabled (80 configurations: 5 $\times$ 4 $\times$ 4 persistence thresholds). Three emergent behaviors were tracked: *curiosity* (observation-dominated baseline), *investigation* (sample or request actions during the disease phase), and *Sherlock Holmes* (a coral-health hidden-cause hypothesis precedes and drives investigation).

The ablation reveals three findings. First, individual EFE terms work as theoretically predicted: info-gain drives curiosity (scaling from 20% at IG = 0.3 to 80% at IG = 3.0), and the contextual viability penalty drives protection (0% at V = 0.5, 100% at V = 5.0). Second, EFE alone is insufficient for the Sherlock Holmes effect—creative abduction is a necessary component, not a convenience. Without it, the system cannot investigate what it does not know it does not know, regardless of parameter tuning (0/90 across all configurations). Third, the cur-

| Emergent behavior | Without abduction | With abduction |
|---|---|---|
| Curiosity | 42/90 (47%) | 0/80 (0%) |
| Investigation | 0/90 (0%) | 80/80 (100%) |
| Sherlock Holmes | 0/90 (0%) | 80/80 (100%) |
| ≥3/5 true causal links | 42/90 (47%) | 20/80 (25%) |

Table 12: Ablation results. Without creative abduction, the EFE info-gain term produces curiosity but cannot produce investigation (no uncertainty to resolve). With creative abduction, the system hypothesizes hidden causes and investigates them, but at the cost of reduced exploration.

rent implementation does not gracefully transition between cognitive modes: configurations that produce investigation sacrifice exploration, and vice versa. The individual terms compose but the mode transitions require further work—this is the primary integration challenge identified by this evaluation.

#### 4.8.3 Resolving the Integration Challenge

The ablation study above identified the central problem: curiosity and investigation compete rather than sequencing. Diagnosis revealed the root cause. At cycle 0, beliefs are initialized to a uniform prior (value 0.5, precision 0.5). When the first observations arrive (cycle 3–4), the prediction errors are large—not because the world is anomalous, but because the agent has not yet calibrated. Without a mechanism to distinguish ignorance from anomaly, the system interprets its own boot-up errors as evidence of hidden causes and immediately generates latent hypotheses for nearly every observable. These hallucinated latents then dominate the EFE landscape, crowding out genuine exploration.

An initial attempt addressed this through two data-driven epistemic gates: an observation coverage requirement (≥60% of beliefs observed at least twice) and a belief stability check (average absolute error $< 0.15$). While effective—achieving 100% curiosity + investigation across all configurations—these gates introduced statutory thresholds that violated the *Bottom-Up* and *Emergent* design principles. Closer analysis revealed that the problem was not in the dynamic equations but in the initial conditions.

**The Boundary Condition Fix.** The surprise formula $S = \frac{1}{2} p\, e^2$ is quadratic in error but *linear* in precision. When precision $p = 0.5$, a prediction error of $e = 0.4$ yields $S = 0.04$—enough to trigger the Hebbian accumulator and abduction generator. When $p = 0.01$, the same error yields $S = 0.0008$—below the cognitive threshold by two orders of magnitude. The system physically *cannot be surprised by things it has no confidence in.* No gate is needed; there is no energy in the signal.

We implement this through two complementary mechanisms, neither of which introduces new thresholds:

**Near-zero initial precision.** Beliefs initialize at $p_0 = 0.01$ instead of 0.5. An *observation-driven precision ratchet* integrates precision upward by a fixed increment ($\Delta p = 0.03$) each time the observable is sampled, independent of prediction accuracy. This reflects "how much data the agent has" rather than "how well it predicted last time." After approximately 15 observations, precision reaches ∼0.46—comparable to the old initial value of 0.5, but earned through data exposure rather than assumption. This creates an emergent "childhood" period where the agent absorbs data without overreacting to calibration errors.

**Metabolic capital for abduction.** The agent maintains a global metabolic capital pool, initialized at zero. Creative abduction requires spending capital ($c_{\text{abd}} = 0.5$). Induction is free. Capital accumulates from successful predictions across all models (+0.02 per correct prediction). This naturally sequences the Peircean triad: the agent must master empirical induction (free) before it can afford theoretical abduction (expensive). The sequencing is an *economic constraint* emerging from the metabolic dynamics, not a gate imposed from outside.

The two mechanisms are complementary. Low initial precision suppresses early surprise signals. Zero initial capital prevents premature abduction even if some signal leaks through. Neither needs to be perfect on its own.

Two additional mechanisms address downstream effects. *Dynamic gestation* assigns abductive passive models a longer protected period (8 cycles) than empirical models (3 cycles), reflecting that hidden-cause hypotheses require more time to manifest confirmable predictions. *Epistemic shielding* dampens the viability penalty for actions that provide information about latent hypotheses linked to the threatened observable:

$$w_{\text{viab}}^{\text{eff}} = w_{\text{viab}} \cdot (1 - \alpha \cdot \min(1, \gamma \cdot \text{IG}_h)), \tag{7}$$

where $\alpha = 0.7$ controls the maximum shielding strength and $\gamma = 8.0$ scales the latent information gain. This permits the system to "think under fire": when viability is threatened and a hypothesis exists about why, the panic reflex is dampened just enough to allow investigation.

| **Metric** | **Baseline** | **Grace Period** | **Epistemic Gates** | **Boundary Fix** |
|---|---|---|---|---|
| Curiosity + Investigation | 0% | 100% | 100% | 100% |
| Sherlock Holmes effect | 100%$^\dagger$ | 40% | 40% | **100%** |
| Avg true causal links (of 5) | 2.0 | 2.8 | 2.6 | **4.0** |
| Avg metabolic deaths | 48 | 37 | 44 | **7** |
| Gold standard* | 0% | 20% | 0% | **100%** |
| Avg phase switches | 0.0 | 1.1 | 1.0 | 1.2 |
| First creative abduction | cycle 3 | cycle 19 | cycle 8 | cycle 38 |
| Design-principle compliant | ✓ | × | ∼ | ✓ |

Table 13: Integration results across 20 parameter configurations (5 info-gain weights × 4 viability weights). Grace Period uses a hardcoded cycle threshold ($\leq 15$). Epistemic Gates uses observation coverage and belief stability thresholds. Boundary Fix uses near-zero initial precision + metabolic capital. *Gold standard = Sherlock Holmes + $\geq$3 true causal links. $^\dagger$Baseline's 100% Sherlock rate is misleading: latent hypotheses are created at cycle 3 from noise, not anomaly detection.

The boundary fix achieves 100% Sherlock Holmes effect and 100% gold standard across all 20 configurations—the only variant to do so (Table 13). Three properties drive this improvement. First, the extended calibration period (first creative abduction at cycle ∼38 vs. cycle 3–19 for other variants) means the agent has built a solid inductive world model *before* attempting abduction. When the disease pathogen appears at cycle 41, the system has well-calibrated beliefs and an established causal structure against which the anomaly is genuinely surprising. Second, the dramatically lower death count (7 vs. 37–48) confirms that hypotheses are created at the right time: because the agent has sufficient context, far fewer hypotheses are spurious. Third, the metabolic capital mechanism naturally sequences the Peircean triad—induction (free, builds world model) precedes abduction (expensive, requires context)—without any explicit sequencing logic.

The precision ratchet dynamics are shown in the detailed trace: average belief precision rises from 0.01 to 0.50 over the first 15 cycles purely from observation exposure, then continues toward 0.95 as predictions improve. This creates an organic transition from data-absorption to anomaly-detection that requires no mode switch.

Crucially, this resolution complies with all five design principles. No enumerated threshold, cycle count, or mode switch is required. The system's "childhood"—the period of low precision and zero capital—emerges from two boundary conditions interacting with the existing dynamic equations. Fix the initial state, not the dynamics.

# 5 Discussion

## 5.1 Design Principles

Five principles guided development:

**Bottom-Up over Top-Down.** Find the invariant (EFE) and let behavior emerge, rather than enumerating cases.

**Symbolic over Statistical.** Use symbols for reasoning; statistics for perception and production only.

**Transparent over Opaque.** Every inference must be traceable, every belief queryable.

**Emergent over Scripted.** Create conditions from which behaviors arise, rather than programming behaviors directly.

**Honest over Fluent.** The system should only claim what it can ground. Uncertainty should be reflected in actual epistemic state.

These principles were not applied post hoc. They actively constrained design decisions—for instance, the principle of emergence prohibited the use of threshold-based mode switching even when it would have been simpler to implement. An initial solution to the integration challenge (epistemic gates based on observation coverage and belief stability) was replaced precisely because it introduced statutory thresholds that violated *Bottom-Up* and *Emergent*, despite being functionally effective. The final resolution—near-zero initial precision and metabolic capital for abduction—achieves stronger results (100% Sherlock Holmes vs. 40%) while introducing zero new thresholds, demonstrating that principled adherence to design constraints can improve rather than limit performance.

## 5.2 Limitations

**Runtime Status.** The MeTTa source is the canonical architecture, written against the MeTTa specification [Potapov, 2021]. Two bugs in the Hyperon 0.2.10 runtime affect native execution.

The first is a **cons-cell evaluation bug**: pattern matching on cons-cell expressions (`($head . $rest)`) does not reduce, breaking recursive list functions including EFE summation over observables. The pure Python MeTTa evaluator (`dagaz_runtime.py`) handles cons-cell destructuring correctly, confirming that the MeTTa source patterns are well-formed and the bug is in the Hyperon evaluator.

The second is a **trie index corruption bug**: after a GroundingSpace's trie index has accumulated approximately 1,000 or more expressions during loading, any subsequent `add-atom` call followed by a `match` query on the newly inserted symbol's trie branch triggers a crash (`trie.rs:179`, unwrap on None). The crash is not caused by file count, symbol collisions, or overlapping type contexts—it is caused specifically by post-load mutation of a trie that has grown past a structural stability threshold. Loading is always safe; dynamic insertion after loading is

not. The corruption is branch-local: querying symbols on other trie branches succeeds even after a corrupting insertion. This bug blocks any MeTTa program that dynamically adds atoms to a large space—exactly the pattern required by a cognitive loop that updates beliefs, records observations, and creates structural hypotheses each cycle.

The architectural resolution is a **three-space separation**: static function definitions reside in `&self` (never mutated after loading, ∼1,600 atoms), dynamic runtime state resides in `&state` (small, freely mutable, ∼50–200 atoms at any time), and immutable semantic knowledge resides in `&ontology` (∼140 atoms). Because `&self`'s trie is never mutated after loading, the bug cannot trigger. Because `&state` stays well below the corruption threshold, dynamic insertion is safe. This separation also produces a cleaner architecture: it enforces the biological distinction between long-term procedural memory (code) and working memory (current state), and it is already implemented in both the MeTTa source and the Python evaluator.

The Python MeTTa evaluator (`dagaz_runtime.py`) provides partial execution of the canonical source. It successfully loads all 22 cognitive modules, computes EFE with correct numerical results, derives affect from prediction errors, selects actions via argmin, and runs complete cognitive cycles producing structured results (total error, affect state, selected action, structure learning state). A stateless orchestrator connects this evaluator to a local language model, running the full perception–cognition–verbalization pipeline. The evaluator does not support MeTTa's full nondeterministic evaluation semantics (bare `match` returns one result rather than all), which means some functions that depend on collecting multiple matches may return partial results. Minimal reproduction cases for both Hyperon bugs are documented in the repository.

**Scale.** Action-space scaling benchmarks (Section 4.7) validate the architecture from 3 to 50 actions and 4 to 24 observables, confirming linear EFE computation, bounded graph density (∼11%), and robust metabolic health (78–89% at 6+ observables). The banded pruning optimization (Section 4.6) addresses pairwise scaling to 1,000 observables. The reef scenario (Section 4.8) demonstrates integrated operation at 14 observables and 8 actions. However, two scale-related questions remain open. First, passive model count growth: at 500 observables, the system generates more spurious hypotheses per cycle, and metabolic pruning must keep pace. Second, the interaction between the band structure and the adaptive metabolic bootstrap has not been tested—if the system adjusts its own rent and reward rates, the $b_{\min}$ cutoff must be rederived. All scaling benchmarks are Python specifications; native MeTTa validation awaits resolution of the Hyperon bugs noted above.

**Meta-parameter Bootstrap.** The metabolic sensitivity analysis (Section 4.5) shows that the healthy basin occupies 67% of parameter space and is governed by a single dimensionless ratio $c_{\text{reward}}/c_{\text{rent}}$. This provides a favorable landscape for the bootstrap mechanism to converge, but the bootstrap itself—which treats metabolic rates as learnable beliefs—has not yet been demonstrated in closed loop.

**Integration Dynamics.** The integration challenge—distinguishing boot-up ignorance from genuine anomaly—was resolved not by adding mechanisms but by correcting boundary conditions (Section 4.8.3). The progression from epistemic gates (statutory thresholds) to the boundary condition fix (correct initial precision plus metabolic capital) illustrates a general principle: when emergent behavior fails, the first diagnosis should be whether the initial conditions are wrong, not whether the dynamics need patching. Near-zero initial precision and zero starting capital produce a "childhood" period in which the system physically cannot hallucinate latent causes, and this period ends naturally as precision integrates from observation exposure and capital accumulates from successful induction. The Sherlock Holmes effect now emerges in 100% of parameter configurations tested, with 4.0 of 5 true causal links discovered and only 7 spurious deaths. The dramatically reduced death count (7 vs. 37–48 under alternative approaches)

confirms that hypothesis timing matters as much as hypothesis generation: creating abductive hypotheses after a solid inductive world model exists means far fewer are spurious. The evaluation uses a single scenario (14 observables, 70 cycles); generalization to other domains has not been verified, and the precision ratchet rate ($\Delta p = 0.03$) and abduction capital cost ($c_{\text{abd}} = 0.5$) are new parameters whose sensitivity has not yet been fully characterized.

### 5.3 Related Work

Active Inference has been implemented in several contexts. Catal et al. [2020] and Millidge [2020] use deep neural networks to approximate generative models, achieving scalability at the cost of transparency. Champion et al. [2022] implement Active Inference agents for discrete decision-making tasks with explicit state spaces. Our work differs in implementing the full reasoning triad (induction, deduction, abduction) as hypothesis generators under metabolic selection, and in using a purely symbolic representation that preserves traceability.

Several mature Active Inference toolkits exist. `pymdp` [Heins et al., 2022] provides a Python library for discrete partially observable Markov decision process (POMDP) agents under Active Inference, with efficient message passing and a modular design. `RxInfer.jl` [Bagaev et al., 2023] implements reactive message-passing inference in Julia, scaling to large factor graphs. Friston's SPM/DEM toolbox [Friston et al., 2017] remains the canonical MATLAB reference implementation. These systems focus on efficient inference within a given generative model. Our architecture addresses an orthogonal problem: how to *discover and maintain* the generative model itself via induction, deduction, and abduction under metabolic selection. The approaches are complementary—a mature inference backend such as `pymdp` or `RxInfer.jl` could serve as the inference engine within a single cycle of our architecture.

Among broader cognitive architectures, SOAR [Laird, 2012] uses a production system with chunking to acquire new rules, ACT-R [Anderson, 2007] models cognition through interacting declarative and procedural modules with Bayesian activation, and LIDA [Franklin et al., 2014] implements a global workspace theory with attention-mediated broadcast. These architectures achieve cognitive breadth through carefully engineered module interactions. Our architecture achieves breadth through a single optimization principle (EFE minimization) applied uniformly across all subsystems, with behavior emerging from the dynamics rather than from inter-module protocols. The metabolic selection mechanism, which provides principled forgetting and bounded growth, has no direct analogue in these architectures—SOAR retains all chunks indefinitely, and ACT-R's base-level decay, while functionally similar, operates on a frequency heuristic rather than predictive accuracy.

The metabolic selection mechanism shares structural similarities with the Attention Economy in the original OpenCog [Goertzel et al., 2010], where atoms compete for attentional resources. Our formulation is simpler (fixed rent and reward rates rather than a market mechanism) and integrated with the Peircean triad.

The RG-flow-inspired planner relates to work on hierarchical planning with adaptive abstraction [Sacerdoti, 1974], but differs in that the abstraction level (beam width) emerges from the EFE landscape rather than being predetermined.

The use of abduction to drive investigation behavior has been explored in explanatory reasoning systems [Josephson and Josephson, 1996], but our contribution is showing that this behavior emerges from EFE minimization without a dedicated investigation module.

The architecture's treatment of the neural-symbolic boundary differs from both pure end-to-end approaches and classical hybrid systems. Rather than using neural components for approximate inference within the reasoning loop [Garcez et al., 2009, Manhaeve et al., 2018] or restricting symbolic processing to a post-hoc verification layer, the system positions neural models exclusively at the perceptual boundary: language models perform dimensionality reduction from continuous input to discrete observations, and the interface is governed by the same precision-weighted Bayesian update that handles all other observation sources (Section 3.8). This is a

division of labor, not a competition: the neural component's contribution is measured by the same prediction error signal that evaluates every other belief in the system. The scaling results of deep learning [Kaplan et al., 2020] are therefore complementary—a more capable perceptual model raises the precision of incoming observations, which tightens the symbolic core's beliefs, but the reasoning architecture itself is unchanged.

# 6 Conclusion

We have presented a cognitive architecture in which a single optimization principle—minimize Expected Free Energy—produces unified perception, action selection, causal reasoning, temporal planning, and honest self-knowledge. Three forms of inference operate as hypothesis generators under uniform metabolic selection. An adaptive planner inspired by Renormalization Group flow reduces planning complexity by up to 80,000× at 50 actions while preserving correct behavior across all seeds tested. A magnitude-banded covariance pruning scheme for the Hebbian suspicion accumulator, with its pruning threshold derived from the metabolic parameters, reduces pairwise scaling from $O(E^2)$ to $O(E \cdot b)$ and achieves 19.4× pair reduction at 1,000 observables with zero significant false negatives. The architecture is implemented in approximately 12,000 lines of MeTTa and validated through executable Python specifications that encode identical logic, a pure Python MeTTa evaluator that executes the canonical source through generic pattern matching, and an end-to-end orchestrator that runs the full cognitive pipeline with a local language model.

The most notable result is the Sherlock Holmes effect: the spontaneous emergence of investigation behavior from the interaction of abductive inference, epistemic value, and EFE minimization. No module was programmed to investigate. The system creates a hypothesis about a hidden cause, recognizes its own uncertainty about that hypothesis, and acts to reduce that uncertainty—all from the same equation that governs routine perception and action. An ablation study confirms that creative abduction is a necessary component: across 90 parameter configurations without abduction, investigation never emerges regardless of tuning.

Initial evaluation revealed an integration challenge: the system could not distinguish its own ignorance (uncalibrated priors) from genuine anomaly (environmental change), causing premature hallucination of latent variables at boot-up. The definitive solution required no new mechanisms—only correct boundary conditions. Near-zero initial precision ($p_0 \approx 0.01$) makes boot-up surprise signals physically negligible, since $S = \frac{1}{2}p \cdot e^2 \approx 0$. An observation-driven precision ratchet integrates confidence from data exposure. A metabolic capital requirement for creative abduction naturally sequences empirical induction before theoretical hypothesis generation. Together, these produce an emergent "childhood" period that ends when the data warrants it, with no gates, thresholds, or cycle counts. The result: 100% Sherlock Holmes effect across all 20 parameter configurations, 4.0/5 true causal links discovered, and only 7 spurious metabolic deaths—compared with 40%, 2.6/5, and 44 under the gate-based approach. This progression from threshold-based patches to correct initial conditions illustrates the design principle that behavior should emerge from dynamics rather than be imposed by rules.

The boundary condition fix introduces two new parameters—the precision ratchet rate and the abduction capital cost—whose sensitivity across domains has not yet been characterized. However, the approach is qualitatively robust: the mechanism operates through the existing surprise equation ($S = \frac{1}{2}p \cdot e^2$) rather than through external thresholds, so any reasonable near-zero initial precision will produce suppression during boot-up, and any positive capital cost will delay abduction relative to induction.

The banded pruning result illustrates a broader architectural property: because the metabolic economy governs both hypothesis selection and the scaling optimization, the system's own parameters define the boundary between computation that matters and computation that can be safely pruned. This suggests that further optimizations—adaptive band resolution, multi-

resolution hashing for temporal patterns—can be derived from the same metabolic foundation rather than introduced as external tuning.

Action-space scaling benchmarks validate that the architecture remains coherent from 3 to 50 actions: EFE computation scales linearly, planning pruning improves with action count, and metabolic selection natively bounds causal graph density to $\sim$11%, solving the structural explosion problem without explicit regularization. Two bottlenecks—a fixed structural budget and a permissive planner noise floor—were resolved by replacing hand-tuned constants with functions of system state.

Characterizing the precision ratchet and capital cost across additional domains, closing the meta-parameter bootstrap loop, and completing native MeTTa execution are the focus of ongoing work.

The source code and all benchmark implementations are available at `https://github.com/Jolnir-Sefir/dagaz`.

## References


Anderson, J. R. *How Can the Human Mind Occur in the Physical Universe?* Oxford University Press, 2007.

Bagaev, D., Podusenko, A., and de Vries, B. RxInfer: A Julia package for reactive real-time Bayesian inference. *Journal of Open Source Software*, 8(84):5161, 2023.

Besold, T. R., d'Avila Garcez, A., Bader, S., Bowman, H., Domingos, P., Hitzler, P., Kühnberger, K.-U., Lamb, L. C., Lowd, D., Lima, P. M. V., de Penning, L., Pinkas, G., Poon, H., and Zaverucha, G. Neural-symbolic learning and reasoning: A survey and interpretation. *arXiv preprint arXiv:1711.03902*, 2017.

Catal, O., Wauthier, S., De Boom, C., Verbelen, T., and Dhoedt, B. Learning generative state space models for active inference. *Frontiers in Computational Neuroscience*, 14:574372, 2020.

Champion, T., Bowman, H., and Grześ, M. Realizing active inference in variational message passing: the outcome-blind fixation of belief. *Neural Computation*, 34(10):2029–2058, 2022.

Franklin, S., Madl, T., D'Mello, S., and Snaider, J. LIDA: A systems-level architecture for cognition, emotion, and learning. *IEEE Transactions on Autonomous Mental Development*, 6(1):19–41, 2014.

Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., and Pezzulo, G. Active inference: A process theory. *Neural Computation*, 29(1):1–49, 2017.

d'Avila Garcez, A., Lamb, L. C., and Gabbay, D. M. *Neural-Symbolic Cognitive Reasoning*. Cognitive Technologies. Springer, 2009.

Goertzel, B., de Garis, H., Pennachin, C., Geisweiller, N., Araujo, S., Pitt, J., Chen, S., Lian, R., Jiang, M., Yang, Y., and Huang, D. OpenCogBot: Achieving generally intelligent virtual agent control and humanoid robotics via cognitive synergy. In *Proceedings of ICAI*, 2010.

Goertzel, B. and Pennachin, C., editors. *Artificial General Intelligence*. Cognitive Technologies. Springer, 2007.

Goertzel, B., Pennachin, C., and Geisweiller, N. *Engineering General Intelligence*. Atlantis Press, 2014.

Heins, C., Millidge, B., Demekas, D., Klein, B., Friston, K., Couzin, I. D., and Tschantz, A. `pymdp`: A Python library for active inference in discrete state spaces. *Journal of Open Source Software*, 7(73):4098, 2022.

Josephson, J. R. and Josephson, S. G. *Abductive Inference: Computation, Philosophy, Technology.* Cambridge University Press, 1996.

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., and Amodei, D. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Laird, J. E. *The Soar Cognitive Architecture.* MIT Press, 2012.

llama.cpp contributors. llama.cpp: LLM inference in C/C++. `https://github.com/ggerganov/llama.cpp`, 2023.

Manhaeve, R., Dumancic, S., Kimmig, A., Demeester, T., and De Raedt, L. DeepProbLog: Neural probabilistic logic programming. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Millidge, B. Deep active inference as variational policy gradients. *Journal of Mathematical Psychology*, 96:102348, 2020.

Millidge, B., Tschantz, A., and Buckley, C. L. Whence the expected free energy? *Neural Computation*, 33(2):447–482, 2021.

Parr, T., Pezzulo, G., and Friston, K. J. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior.* MIT Press, 2022.

Peirce, C. S. *Collected Papers of Charles Sanders Peirce*, volumes 1–6. Harvard University Press, 1935. Edited by C. Hartshorne and P. Weiss.

Potapov, A. MeTTa specification. Technical report, SingularityNET Foundation, 2021.

Sacerdoti, E. D. Planning in a hierarchy of abstraction spaces. *Artificial Intelligence*, 5(2):115–135, 1974.

Wilson, K. G. and Kogut, J. The renormalization group and the $\epsilon$ expansion. *Physics Reports*, 12(2):75–199, 1974.